



TRACKIFY:
DIGITAL PLATFORM FOR STUDENT ASSIGNMENT AND
PROJECT MANAGEMENT

Phase 1-Project Proposal and Planning

Group Mario

Section 03

Nur Dania Halisa binti Mohd Hazri, A25CS0306

Nur Amarah Auni binti Mazlan, A25CS0302

Nurin Batrisyia Husna binti Mohd Hazry, A25CS0328

Siti Nur Alysha binti Suhaimi, A25CS0354

Sofea binti Mohammad Hisyamuddin, A25CS0356

Faculty of Computing, Universiti Teknologi Malaysia

SECD2613 System Analysis and Design

Associate Professor Dr. Norasnita Ahmad

April 16, 2026

	Content	Page
	1.0 Introduction	3
	2.0 Background Study	3
	3.0 Problem Statements 3.1 Busy Schedules and Time Management Issues 3.2 Communication Gaps 3.3 Missed Deadlines 3.4 Difficulty with e-Learning Systems 3.5 Lack of Assignments progress Monitoring	4
	4.0 Proposed Solutions 4.1 Smart Reminders 4.2 Assignment Progress Tracking 4.3 Group Assignment Tracking 4.4 Priority Ranking System Mode 4.5 Feasibility Study 4.6 Cost-Benefit Analysis (CBA)	4 5 6
	5.0 Objectives	7
	6.0 Scope of the Project	7
	7.0 Project Planning 7.1 Human Resource 7.2 Work Breakdown Structure (WBS) 7.3 Pert Chart 7.4 Gantt Chart	8 9 10 11
	8.0 Benefits and Overall Summary of Proposed System	11
	9.0 GitHub Information	12

1.0 Introduction

Assignment tracking plays an important role in helping students manage their academic workload effectively. In UTM, students are often required to handle multiple assignment and group projects at the same time, and each have different deadlines and requirements. This makes it important for students to have a clear and organized way to monitor their progress and ensure that their tasks are completed on time. Without a proper system, students may feel overwhelmed, forget important deadlines, or struggle to prioritize their tasks efficiently. In addition, managing group projects can become challenging when there is a lack of coordination and communication among team members. Therefore, having an effective assignment tracking method can help improve time management, reduce stress, and enhance overall academic performance.

2.0 Background Study

Students in UTM rely on various of tools when it comes to keeping track of their assignment progress or making the project deadlines. Some of the students also only depend on their friends or lecturer to remind them when the due of the project or assignments are approaching. These methods are commonly used as they are simple and familiar among students but it lacks efficiency as different tools will scatter all the assignment information across different platform.

In addition, group assignments usually require continuous communication and coordination among members. Students rely on group chats to discuss tasks distribution, share updates, and remind each other about deadlines. While this method allows quick communication, the information shared can be easily lost or overlooked as more messages are sent over time.

3.0 Problems Statements

3.1 Busy Schedules and Time Management Issues

Packed timetables and involvement in university programs or associations are challenging for students to allocate time effectively.

3.2 Communication Gaps

For group projects, there may be lack of communication between the team members regarding project progress for divided tasks.

3.3 Missed Deadlines

Students may forget assignment deadlines or struggle with tracking multiple assignments from different classes.

3.4 Difficulty with e-Learning Systems

The system only shows the deadlines when students are logged in which makes it difficult and inconvenience in some cases. It also does not provide a progress tracker for progress monitoring.

3.5 Lack of Assignments progress Monitoring

Students lack proper tools to visualize progress and track task completion.

4.0 Proposed Solutions

4.1 Smart Reminders

Popped up notification on the user's smartphone or tablet. Student does not have to login into e-learning and find the assignment deadlines for each subject manually.

4.2 Assignment Progress Tracking

Displaying the percentages of work done with a to do list.

4.3 Group Assignment Tracking

Team members will be able to communicate with each other more efficiently and see their real time progress.

4.4 Priority Ranking System Mode

Assignment or project with the highest priority, whether it be the closest due date and such, will automatically move to the top of the assignment list.

4.5 Feasibility Study

4.5.1 Technical Feasibility

The development of the Assignment Tracker application is technically feasible as it relies on widely available technologies. The system can be developed using platforms such as Figma. These tools are accessible and suitable for student-level development. Additionally, features such as notifications, task management, and data storage are commonly supported, reducing implementation complexity. Therefore, the technical requirements are achievable within the current technological environment.

4.5.2 Operational Feasibility

The proposed system is operationally feasible as it directly addresses common challenges faced by students, such as forgetting deadlines and poor task organization. The application is intended to be user-friendly, with a simple interface that allows users to easily input, update, and monitor their assignments. Features such as reminders, priority setting, and progress tracking enhance usability and effectiveness. As the system aligns with user needs and requires minimal training, it is expected to be widely accepted and effectively utilized.

4.5.3 Economic Feasibility

Criteria	Year 0	Year 1	Year 2	Year 3
1. Costs				
A. Development Costs				
Hardware	2,700	-	-	-
Software	450	-	-	-
Training	270	-	-	-
Total Development Cost	3,420			
2. Production Costs				
Advertisement	-	1,600	1,690	1,965
Maintenance	-	4,050	4,253	5,730
Salary	-	19,800	20,790	21,830
Total Production Costs (Annual)	-	25,650	27,203	29,545
Present Value (PV) $PV = \text{Payment} \times 1 / (1 + c)^n$	-	23,318	22,483	22,198
Accumulated Costs	-	26,738	49,221	71,419
2. Benefits				
Time saving	-	12,000	12,600	13,230
Reduced missed deadlines	-	9,600	10,080	10,584
Productivity improvement	-	8,400	8,620	9,261
Total Benefits (Annual)	-	30,000	31,500	33,075
Present Value (PV) $PV = \text{Payment} \times 1 / (1 + c)^n$	-	27,273	26,033	24,850
Accumulated Benefits	-	27,273	53,306	77,886
Gain or Loss	-	535	4,085	6,467
Profitability Index	1.89			

The profitability index (PI) is 1.89, which is greater than 1, indicating that the project is financially feasible and represents a good investment.]

5.0 Objectives

- 1) Provide Centralized Assignment Management
 - Allow students to record, monitor, and update assignment progress.
- 2) Enhance Time Management
 - Reduce the risk of missed submissions through smart reminder notifications.
- 3) Support Collaborative Tasks
 - Strengthen teamwork through built-in collaboration tools.
- 4) Provide Task Prioritization
 - Help students focus on the most urgent or important tasks.
- 5) Ensure Accessibility and Usability
 - Deliver a user-friendly web interface that is easy to navigate anytime.

6.0 Scope of the Project

What will be Included	What will not be Included
Assignment Progress Tracking (Monitor deadlines and tasks)	No Automated Grading (Manual input required)
Smart Reminder (Automated notification before deadlines)	No Plagiarism Detector
Group Chat or Discussion (Centralized team communication)	
Priority Ranking System (Sort Tasks by Importance)	

7.0 Project Planning

7.1 Human Resource

A. Overview

This project requires several human resources to ensure the successful development and implementation of the system. Each role contributes to different aspects such as design, development and testing.

B. Roles and Responsibilities

1) Project Manager

- Plans the project timeline
- Manages tasks and ensures deadlines are met'
- Coordinate team members

2) Web Developer

- Build the system
- Implement features like assignment submission and notifications

3) UI/UX Developer

- Designs user-friendly interface
- Ensure the system is easy to use
- Choose layouts and structures

4) Data Analyst

- Analyzes assignment data collected in the system
- Ensure data is useful for users
- Identifies patterns like late submissions or missed deadlines

5)System Analyst

- Studies of user needs and system requirements
- Acts as a bridge between user needs and system design

7.2 Work Breakdown Structure (WBS)

1) Project Initiation

- 1.1 Identify Problems
- 1.2 Define project scope
- 1.3 Define project objectives
- 1.4 Conduct feasibility study

2) Planning

- 2.1 Develop project plan
- 2.2 Define budget
- 2.3 Set timeline and milestones
- 2.4 Resource allocation

3) Requirement Analysis

- 3.1 Conduct user research
 - 3.1.1 Student survey
 - 3.1.2 Lecture feedback
- 3.2 Analyse current system issues
- 3.3 Define functional requirements
 - 3.3.1 Login system
 - 3.3.2 Assignment tracking
 - 3.3.3 Deadline notification
 - 3.3.4 Progress update
 - 3.3.5 Priority ranking system
- 3.4 Define non-functional requirements
 - 3.4.1 Security
 - 3.4.2 Usability
 - 3.4.3 Performance
 - 3.4.4 Reliability

4) System Design

- 4.1 UI/UX wireframes
- 4.2 Database design
- 4.3 System architecture
- 4.4 API and integration planning
- 4.5 Security framework

5) System development

- 5.1 Frontend development
- 5.2 Backend development
- 5.3 Database implementation
- 5.4 System integration

6) Testing and QA

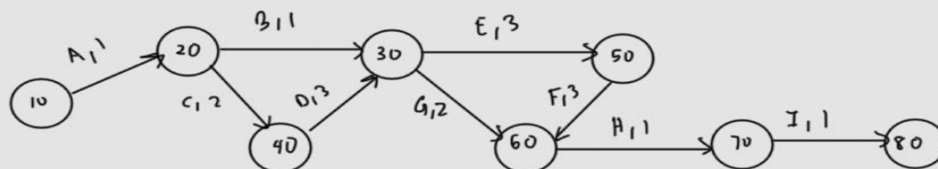
- 6.1 Unit testing
- 6.2 System testing
- 6.3 Integration testing
- 6.4 User acceptance testing
- 6.6 Performance testing
- 6.7 Bug fixing

7) Deployment and maintenance

- 7.1 Hosting setup (cloud/server)
- 7.2 Configure SSL and Security Policies
- 7.3 System Deployment
- 7.4 User training
- 7.5 Ongoing bug fix and support
- 7.6 System updates
- 7.7 Feature enhancements
- 7.8 Database backup and monitoring

7.3 Pert Chart

Activity	Predecessor	Duration
A. Project initiation	None	1
B. Define project objectives	A	1
C. Planning	A	2
D. Requirements Analysis	C	3
E. System design	B, D	3
F. System development	E	3
G. Testing and QA	B, D	2
H. Deployment and maintenance	G, F	1
I. System updates	H	1



Path 1: A-B-E-F-H-I Length = 1+1+3+3+1+1 = 10 weeks
Path 2: A-C-D-G-H-I Length = 1+2+3+2+1+1 = 10 weeks
Path 3: A-B-G-H-I Length = 1+1+2+1+1 = 6 weeks
Path 4: A-C-D-E-F-H-I Length = 1+2+3+3+3+1+1 = 11 weeks

∴ Since, the critical path is the longest path through diagram, path 4. A-C-D-E-F-H-I, is the critical path for project.

7.4 Gantt Chart

PROJECT SCHEDULING GANTT CHART

Task Name	Week													
	3	4	5	6	7	8	9	10	11	12	13	14		
Project Initiation														
Define Project Objectives														
Planning														
Requirement Analysis														
System Design														
Sysetm Development														
Testing and QA														
Deploement and Maintenance														
System Updates														

8.0 Benefits and Overall Summary of Proposed System

- 1) Improve time management and prioritization of tasks.
- 2) Reduced stress by providing clear visibility of upcoming deadlines.
- 3) Enhance collaboration through shared tasks boards and GitHub integration.
- 4) Contribution to academic performance by minimizing missed submission and promoting teamwork.

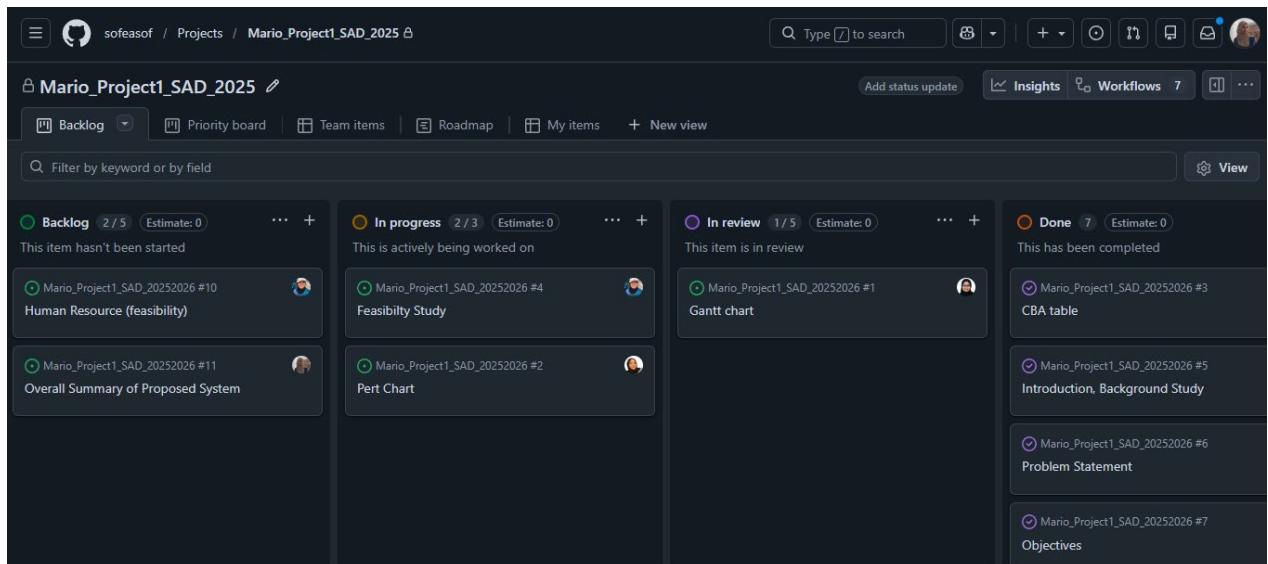
Trackify is designed to make student life a bit easier and not too overwhelming. Instead of trying to keep track of deadlines on different platforms or relying on friends' reminders, Trackify brings everything into one place. With smart reminders, progress trackers, and group collaboration tools, students can stay on top of their work without the constant stress of forgetting something important. The priority ranking system ensures that the most urgent tasks rise to the top, so it helps students focus on what is urgent.

The system is easy to build, and simple to use, making it realistic for student-level development. More importantly, it directly deals with common challenges like missed deadlines, scattered communication, and poor visibility of progress. By giving students, a clear view of their workload and a simple way to coordinate with teammates, Trackify not only improves time management but also reduces stress and strengthens collaboration. In short, it's a tool that helps students feel more in control, more organized, and better prepared to do well in their academic journey.

9.0 GitHub Information

URL:

- https://github.com/sofeasof/Mario_Project1_SAD_20252026 (Repository)
- <https://github.com/users/sofeasof/projects/1> (Project)



Snapshot of the repository/project during project planning process.